

Exercise 5.1 Consider a code that stores $k = 4$ bits x_1, x_2, x_3 and x_4 in $n = 8$ bit codewords by computing the parities $x_1 \oplus x_2$, $x_3 \oplus x_4$, $x_1 \oplus x_3$ and $x_2 \oplus x_4$ and storing them together with the original bits.

- Give the codewords for this code and compute the minimal distance. How many errors can it detect and correct?
- Is it a linear code? If so, compute matrices G and H .
- Use the Hamming bound to determine if this code is perfect or not.
- Construct the dual code for this code.

Solution:

- The codewords are:

codeword	$d_{\min}$	codeword	$d_{\min}$	codeword	$d_{\min}$	codeword	$d_{\min}$
00000000	3	00010101	3	00100110	3	00110011	3
01001001	3	01011100	3	01101111	3	01111010	3
10001010	3	10011111	3	10101100	3	10111001	3
11000011	3	11010110	3	11100101	3	11110000	3

Let $2t + 1 = 3$ and then $t = 1$. The code can detect up to 2 bit flip errors, correct up to 1 bit flip errors and correct up to 2 erasures.

- It is a linear code.

$$G = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \end{bmatrix}, \quad H = \begin{bmatrix} 1 & 1 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 & 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 & 0 & 0 & 0 & 1 \end{bmatrix}.$$

- Hamming bound: $d = 3$, $n = 8$

$$16 = |C| < \frac{2^n}{\sum_{i=0}^{\lfloor (d-1)/2 \rfloor} \binom{n}{i}} = \frac{2^8}{9}.$$

So this code is not perfect.

- For the dual code: $G^\perp = H^T$ and $H^\perp = G^T$, that is

$$G^\perp = \begin{bmatrix} 1 & 0 & 1 & 0 \\ 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \quad H^\perp = \begin{bmatrix} 1 & 0 & 0 & 0 & 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 & 1 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 & 1 & 0 & 1 \end{bmatrix}.$$

Exercise 5.2 Construct the dual code of the Hamming code. What is its rate and code distance?

Solution: For the dual of the Hamming code, we have

$$G = H_{\text{Hamming}}^T = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad H = G_{\text{Hamming}}^T = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 1 & 1 \\ 0 & 1 & 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 & 1 & 1 & 1 \end{bmatrix}.$$

The code distance is equal to the minimal Hamming weight among the codewords, which is 4. The rate is

$$R(C) = \frac{\log(|C|)}{\log(|\Sigma^7|)} = \frac{\log(2^3)}{\log(2^7)} = \frac{3}{7}.$$

Exercise 5.3 Show the following properties of a (binary) linear $[n, k]$ -code C .

- If H is the parity check matrix of C , then $d(C)$ equals the minimal number of columns of H that are linearly dependent.
- Prove that (after permuting the coordinates if necessary) C has a generator matrix of the form $G = [I_k \ G']^T$ where I_k is the $k \times k$ identity matrix, and where G' is some $k \times (n - k)$ matrix.

Solution:

- Let $H = [h_1^T, \dots, h_n^T] \in \{0, 1\}^{(n-k) \times n}$, where $h_i^T \in \{0, 1\}^{(n-k) \times 1}$. Let $c = [c_1, \dots, c_n]^T$ be the codeword with minimal Hamming weight. Since H is the parity check matrix of C , then

$$0 = Hc = \sum_{i=1}^n h_i^T c_i = \sum_{i: c_i \neq 0}^n h_i^T = 0.$$

Thus, $d(C)$ is the minimal number of columns of H that are linearly dependent: if there was a smaller number of linearly dependent columns in H , we could find a $\tilde{c}$ with Hamming weight less than that of c , for which $H\tilde{c} = 0$, a contradiction.

- Since the dimension of C is k , there must exist k linear independent vectors $\mathbf{v}_1, \dots, \mathbf{v}_k$ such that $C = \text{span}(\{\mathbf{v}_1, \dots, \mathbf{v}_k\})$. By definition, $G := [\mathbf{v}_1, \dots, \mathbf{v}_k] \in \mathbb{F}^{k \times n}$ is a generator matrix of C . We argue that, allowing swapping the rows of G (only swapping) and columns reductions (scaling one column of G by a non-zero scalar and adding linear combinations of other columns onto this column), we can rewrite G into the canonical form described in the question. Before proving this to be doable, we argue that the matrix after above operations is still a valid generator matrix of C (or its equivalent) after each action:

- If the action taken is a row swap, the matrix is a generator matrix of C with the corresponding coordinates swapped.
- If the action taken is adding a column to another column, one can check that the new column is still linearly independent. Additionally, since the new set of vectors (consisting of all the columns now) is a linear combination of the original basis, they span a subspace of C with same dimension, and thus spans C . In other words, the new columns are again a basis of C .

Now, we prove that these two operations can transfer G into a canonical form. We prove this by induction:

i.) Firstly, after a finite number of the allowed operations, we can turn the matrix G into

$$\begin{pmatrix} 1 & 0 & \cdots & 0 \\ v_{1,2} & v_{2,2} & \cdots & v_{k,2} \\ \vdots & \vdots & \ddots & \vdots \\ v_{1,n} & v_{2,n} & \cdots & v_{k,n} \end{pmatrix}.$$

This is done by scaling the first column by $g_{1,1}^{-1}$ and add $-g_{j,1}/g_{1,1} \cdot \mathbf{g}_1$ to the j -th column ($j = 2, \dots, k$), where we denote the i -th column of G by $\mathbf{g}_i$ and ℓ -th entry of $\mathbf{g}_i$ by $g_{i,\ell}$. In case when $g_{1,1} = 0$, we can swap the first row of G with ℓ -th row of G where $g_{1,\ell} \neq 0$. This is doable since $\mathbf{g}_1 \neq 0$.

ii.) Assume after i steps where $i < k$, we can rewrite G as

$$\begin{pmatrix} 1 & \cdots & 0 & 0 & \cdots & 0 \\ \vdots & \ddots & \vdots & \vdots & \ddots & 0 \\ 0 & \cdots & 1 & 0 & \cdots & 0 \\ v_{1,i+1} & \cdots & v_{i,i+1} & v_{i+1,i+1} & \cdots & v_{k,i+1} \\ \vdots & \ddots & \vdots & \vdots & \ddots & \vdots \\ v_{1,n} & \cdots & v_{i,n} & v_{i+1,n} & \cdots & v_{k,n} \end{pmatrix}.$$

Next, scale the $i+1$ -th column by $(v_{i+1,i+1})^{-1}$. In case $v_{i+1,i+1} = 0$, we can first swap the $i+1$ -th row with the j -th row where $v_{i+1,j} \neq 0$, $j > i+1$. This is doable since the dimension of the code being k ensures that not all of $\{v_{i+1,i+1}, \dots, v_{i+1,n}\}$ are zero. The matrix will then look like this

$$\begin{pmatrix} 1 & \cdots & 0 & 0 & \cdots & 0 \\ \vdots & \ddots & \vdots & \vdots & \ddots & 0 \\ 0 & \cdots & 1 & 0 & \cdots & 0 \\ v'_{1,i+1} & \cdots & v'_{i,i+1} & 1 & \cdots & v'_{k,i+1} \\ \vdots & \ddots & \vdots & \vdots & \ddots & \vdots \\ v'_{1,n} & \cdots & v'_{i,n} & v'_{i+1,n} & \cdots & v'_{k,n} \end{pmatrix}.$$

Now, for each $j \neq i+1$ where j is between 1 and k , add $-v'_{j,i+1} \cdot \mathbf{v}'_{i+1}$ to the j -th column. Here, $\mathbf{v}'_{i+1}$ denotes the $i+1$ -th column. This will turn the matrix into this:

$$\begin{pmatrix} 1 & \cdots & 0 & 0 & \cdots & 0 \\ \vdots & \ddots & \vdots & \vdots & \ddots & 0 \\ 0 & \cdots & 1 & 0 & \cdots & 0 \\ 0 & \cdots & 0 & 1 & \cdots & 0 \\ \vdots & \ddots & \vdots & \vdots & \ddots & \vdots \\ v''_{1,n} & \cdots & v''_{i,n} & v'_{i+1,n} & \cdots & v''_{k,n} \end{pmatrix}.$$

iii.) By i) and ii), we know the above process can continue until $i = k$. In the case when $i = k$, we have

$$G \sim \begin{pmatrix} I_k \\ G' \end{pmatrix}$$

for some $G' \in \mathbb{F}^{k \times (n-k)}$.

Exercise 5.4 Assume you are given a linear code by its $n \times k$ generator matrix G or by its $(n-k) \times n$ parity-check matrix H . Which of the below operations can cause a reduction or increase in the minimum code distance?

- a) *Exchanging two rows of G .*
- b) *Exchanging two rows of H .*
- c) *Exchanging two columns of G .*
- d) *Exchanging two columns of H .*
- e) *Deleting a row of G .*
- f) *Deleting a row of H .*
- g) *Deleting a column of G .*
- h) *Deleting a column of H .*
- i) *Adding one column of H to another column of H .*

Solution:

- a.) No.
- b.) No.
- c.) No.
- d.) No.
- e.) Yes. Since the length n of the codeword is shortened and the minimum weight $d \leq n - k + 1$. Removing a bit from the codewords can make them more similar.
- f.) Yes, n is unchanged but $n - k$ is reduced and thus k increases. This may cause a reduction in d , since we effectively add codewords to the code.
- g.) Yes. We reduce the codespace, which might increase the minimal distance (e.g., for the repetition code).
- h.) No. Say we remove column i from H . Then we can obtain the new codewords by keeping all original codewords which have an entry 0 at position i , which we remove from the vector. The Hamming distance does not change.
- i.) Yes. Take the Hamming code, where we add the first column of H to the second column, resulting in $\tilde{H}$. Note that the second column of $\tilde{H}$ is equal to the third column, such that $x = [0, 1, 1, 0, 0, 0, 0]$ is in the new code, which has Hamming weight 2, which upper bounds the distance of the new code. Thus, the distance reduced compared to the original code.

Exercise 5.5 Consider the field F_{2^n} . For every $s \in F_{2^n}$ with $s \neq 0$, define the function

$$f_s : \{0, 1\}^n \rightarrow \{0, 1\}^\ell, \quad x \mapsto x \cdot s \pmod{2^\ell},$$

where the multiplication is with regards to the field F_{2^n} . Show that the family $\{f_s\}_s$ is a two-universal family of hash functions.

Solution: Suppose that

$$f_s(x) = f_s(x').$$

Then we have

$$x \cdot s = x' \cdot s \pmod{2^\ell},$$

and thus

$$(x - x') \cdot s = 0 \pmod{2^l}.$$

Suppose that

$$x - x' \neq 0.$$

Then by the property of F_{2^n} , $(x - x') \cdot s$ generates $F_{2^n} \setminus \{0\}$ as we iterate s in $F_{2^n} \setminus \{0\}$ (for all $a \in F_{2^n}$ and $a \neq 0$, for all $b \in F_{2^n}$ and $b \neq 0$, you can always find $c = a^{-1} \cdot b \in F_{2^n}$ and $c \neq 0$ such that $a \cdot c = b$). Therefore, $(x - x') \cdot s = 0 \pmod{2^l}$ but $x - x' \neq 0$ both holds only when the lowest l bits of $(x - x') \cdot s$ are 0. By iterating s in $F_{2^n} \setminus \{0\}$, there are only $2^{n-l} - 1$ values of s satisfying the condition. Therefore

$$P[f_s(x) = f_s(x')] = \frac{2^{n-l} - 1}{2^n} = \frac{1}{2^l} - \frac{1}{2^n} \leq \frac{1}{2^l}$$